EFC Ontological Foundations v1.0

The Co-Primary Structure

Energy-Flow and Entropy as Differentiated Aspects of Pre-Geometric Potential

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Abstract

This document establishes the ontological foundations for Energy-Flow Cosmology (EFC). Through cross-domain coherence analysis drawing on Wheeler (information ontology), Verlinde (entropic gravity), and Prigogine (dissipative structures), we identify the **co-primary structure** as the most logically consistent foundation: Energy-flow (EF) and entropy (S) are not separate primordial entities but differentiated aspects of a single pre-geometric potential. This resolves the circularity problem inherent in monistic ontologies and provides EFC with a non-ad-hoc metaphysical grounding compatible with EBE, RCMP, and Core Lock.

Keywords: ontology, co-primacy, pre-Planck physics, entropy, energy-flow, emergence, EFC

Epistemic Status: Layer C (Methodological). Coherence argument, not empirical claim. Provides logical foundation for EFC architecture.

1 The Circularity Problem

Any emergent cosmology faces a fundamental question: What is ontologically primary? In theories where gravity, spacetime, or structure are emergent phenomena, something must be assumed to exist before them. This creates potential circularity.

1.1 The Monistic Trap

Consider two natural candidates for ontological primacy:

Energy-Flow Primacy (H1)

If energy-flow is primary, we must ask: *What flows, and through what medium, before structure exists?* Flow presupposes a substrate and directionality. Without spacetime, these concepts become circular.

Entropy Primacy (H2)

If entropy is primary, we must ask: *Entropy of what? What states are being counted before there are states to count?* Entropy is defined over microstates. Without structure, there are no microstates.

Both monistic positions generate hidden circularity. They require the very structure they claim to explain as a prerequisite for their primary entity.

1.2 The Grid-First Fallacy

One might attempt to resolve this by positing a primordial grid or lattice structure (H4). But this merely displaces the problem: What is the grid made of? What sustains it? This introduces unexplained metaphysical furniture—physics before physics.

2 Cross-Domain Evidence

Before proposing a resolution, we examine how related theoretical frameworks handle the primacy question.

Framework	**Key Insight**	**Implication for EFC**
Wheeler (1989)	"It from Bit"—information is fundamental to physical existence	Information/entropy has ontological weight, not merely descriptive
Verlinde (2010)	Gravity emerges from entropy gradients on holographic screens	Entropy can drive dynamics without being temporally prior to them
Prigogine (1977)	Dissipative structures emerge from energy flow in open systems	Flow creates structure; structure enables flow—mutual dependence
Holographic Principle	Information on boundaries encodes bulk physics	Spacetime may be emergent from more fundamental information
Pre-Planck Physics	Below Planck scale, spacetime distinctions dissolve	The very categories of "state" and "process" may be emergent

Table 1: Cross-domain theoretical frameworks and their implications for EFC ontology.

A pattern emerges: The most successful approaches treat information/entropy and dynamics/flow as co-arising rather than one preceding the other.

3 The Co-Primary Resolution

3.1 Core Proposal

Energy-flow (EF) and entropy (S) are not separate primordial entities. They are two aspects of a single, undifferentiated pre-geometric potential that differentiates at the Planck transition.

This resolves the circularity problem: Neither EF nor S is temporally or ontologically prior. The question "which came first?" is ill-posed, like asking which came first, the inside or outside of a sphere.

Phase	Characterization	Ontological Status
Pre-Planck	$t < t_{\text{Planck}}$. No spacetime. EF and S are undifferentiated aspects of one potential.	Distinctions not applicable. The concepts "state" and "process" have not yet emerged.
Planck Transition	Symmetry breaking. Differentiation emerges.	S manifests as "what the system IS" (state). EF manifests as "what the system DOES" (dynamics).
Post-Planck	Standard physics. Grid emerges as medium. Structure forms through S-EF interplay.	EFC phenomenology applies. EBE regimes are well-defined. Core Lock dynamics operative.

Table 2: The three phases of ontological differentiation.

Criterion	EF-Primary (H1)	S-Primary (H2)	Co-Primary (H3)
Circularity	FAILS	FAILS	PASSES
Pre-Planck coherence	FAILS	FAILS	PASSES
Wheeler compatibility	Partial	Strong	Strong
Verlinde compatibility	Partial	Strong	Strong
Prigogine compatibility	Strong	Partial	Strong
EFC integration	Partial	Partial	Complete

Table 3: Comparison of ontological hypotheses against key criteria.

3.2 The Three Phases

3.3 Why Co-Primary Succeeds Where Monisms Fail

4 Integration with EFC Architecture

The co-primary ontology provides a coherent foundation for all layers of the EFC framework:

4.1 Core Lock (Dynamical Engine)

The RG-flow parameter $f(S)$ in Core Lock describes *the mechanism of differentiation*. The constant beta-function ($\beta = -1/\Delta S$) uniquely determines exponential relaxation from the undifferentiated pre-Planck state to differentiated post-Planck regimes.

The accumulated phase $\Phi(S) = \int dS'/f(S')$ represents the total information length, an invariant that bridges the undifferentiated potential to observable quantities through projection laws.

4.2 EBE (Epistemic Framework)

The S-axis in EBE is now **ontologically grounded**. The regime boundaries (Low-S, Mid-S, High-S) are not arbitrary classifications but reflect the degree of S-EF differentiation. Near singularity ($S \to 0$), the distinction weakens; in structured regimes ($S > 0.5$), the distinction is sharp.

The Pre-Physical Boundary (L0) corresponds exactly to the Planck transition where differentiation becomes applicable.

4.3 RCMP (Measurement Protocol)

Driver-proximity becomes deeper: The principle that measurement validity depends on regime is explained by the degree of S-EF differentiation. In regimes where the differentiation is incomplete or blurred, standard measurement assumptions break down.

4.4 EFC Phenomenology

Emergent time, structure formation, and regime-dependent physics all follow from the S-EF interplay in post-Planck space. The Grid is not ontologically primary but is the first emergent structure where differentiated S and EF can interact.

5 Epistemic Status and Limitations

5.1 What This Document Claims

- ✓ The co-primary structure is the most logically coherent ontology for EFC, eliminating circularity problems.
- ✓ It integrates consistently with all EFC sub-frameworks (EBE, RCMP, Core Lock).
- ✓ It aligns with established theoretical frameworks (Wheeler, Verlinde, Prigogine).

5.2 What This Document Does NOT Claim

- × This is not an empirical claim. Pre-Planck physics is by definition beyond direct observation.
- × This does not provide a concrete microphysical mechanism for the Planck transition.
- × The "undifferentiated potential" is a conceptual placeholder, not a fully specified mathematical object.

5.3 Falsifiability

While the ontological foundation itself is not directly testable, its *consequences* are. Core Lock predicts that if any fundamental constants vary, they must exhibit correlated behavior with constant log-log slope. This correlation signature is the empirical bridge between ontology and observation.

6 Conclusion

The co-primary ontology resolves a fundamental problem in emergent cosmology: how to ground a framework without hidden circularity or unexplained metaphysical assumptions.

By treating energy-flow and entropy as **differentiated aspects of a single pre-geometric potential**, rather than competing candidates for primacy, EFC gains a consistent ontological foundation that:

- Eliminates the chicken-and-egg problem of monistic approaches
- Provides logical grounding for the S-axis and regime structure of EBE
- Explains why Core Lock dynamics can operate along the entropy parameter
- Maintains intellectual humility about pre-Planck physics while providing conceptual coherence

This document establishes Layer 0 of the EFC architecture, the ontological foundation upon which Core Lock (dynamics), EBE (epistemology), RCMP (methodology), and EFC phenomenology rest.

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Version History

Version	**Date**	**Changes**
1.0	February 2026	Initial specification. Co-primary ontology established.